

No. of Functional Features Included in the Solution

Date	05 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Interface	Accepts soil nutrients and weather parameters from the user.	User Interface	Done	High
2	Data Validation	Validates input values before processing.	Backend	Done	Medium
3	Crop Recommendation	Predicts the most suitable crop using the trained ML model.	ML Prediction Engine	Done	High
4	Machine Learning Model	Loads the trained model and performs prediction.	Model Module	Done	High
5	Flask Web Application	Provides a browser-based interface for users.	Flask Backend	Done	Medium
6	Dataset Management	Uses the Kaggle crop recommendation dataset for training and prediction.	Data Module	Done	Medium
7	Prediction Result Display	Displays the recommended crop to the user.	Frontend	Done	High
8	Model Storage	Saves and loads the trained ML model using Joblib (.pkl).	Storage Module	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Input Form, Prediction Result Display
2	Backend / Logic	3	Data Validation, Flask Processing, ML Prediction
3	Database / Storage	2	Crop Dataset, Trained Model (.pkl)
4	API / Integration	1	Flask Application Integration
5	Security / Authentication	1	Input Validation and Secure Data Handling